

Graduation Project 2025

Design of Fire Fighting System for Hospital

PROJECT TITLE :

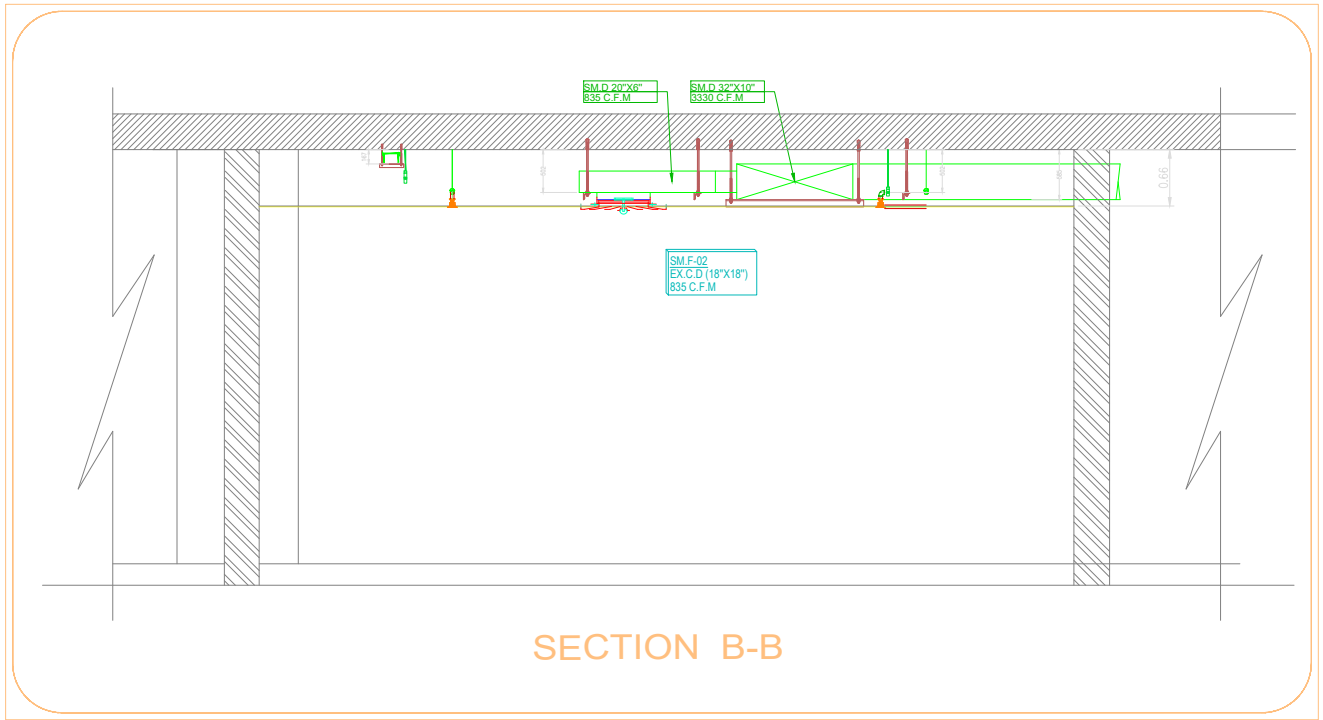
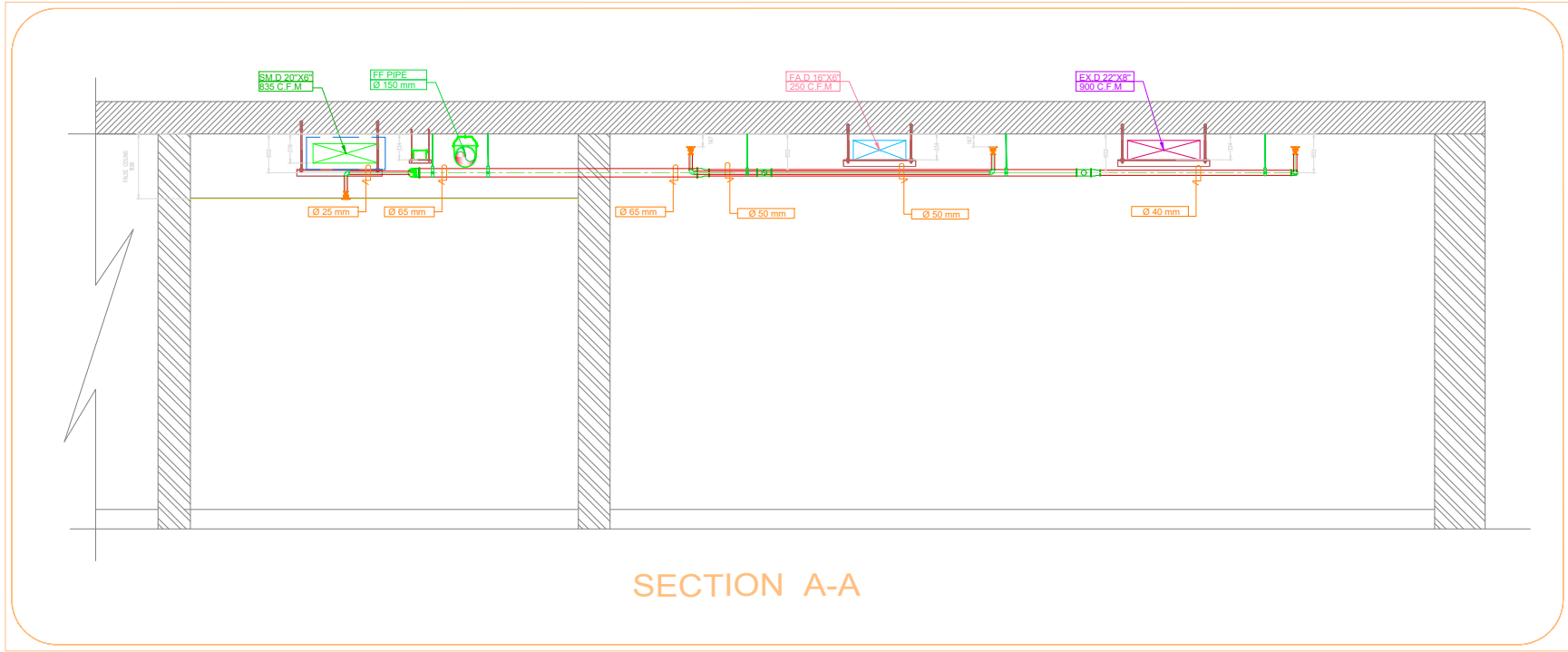
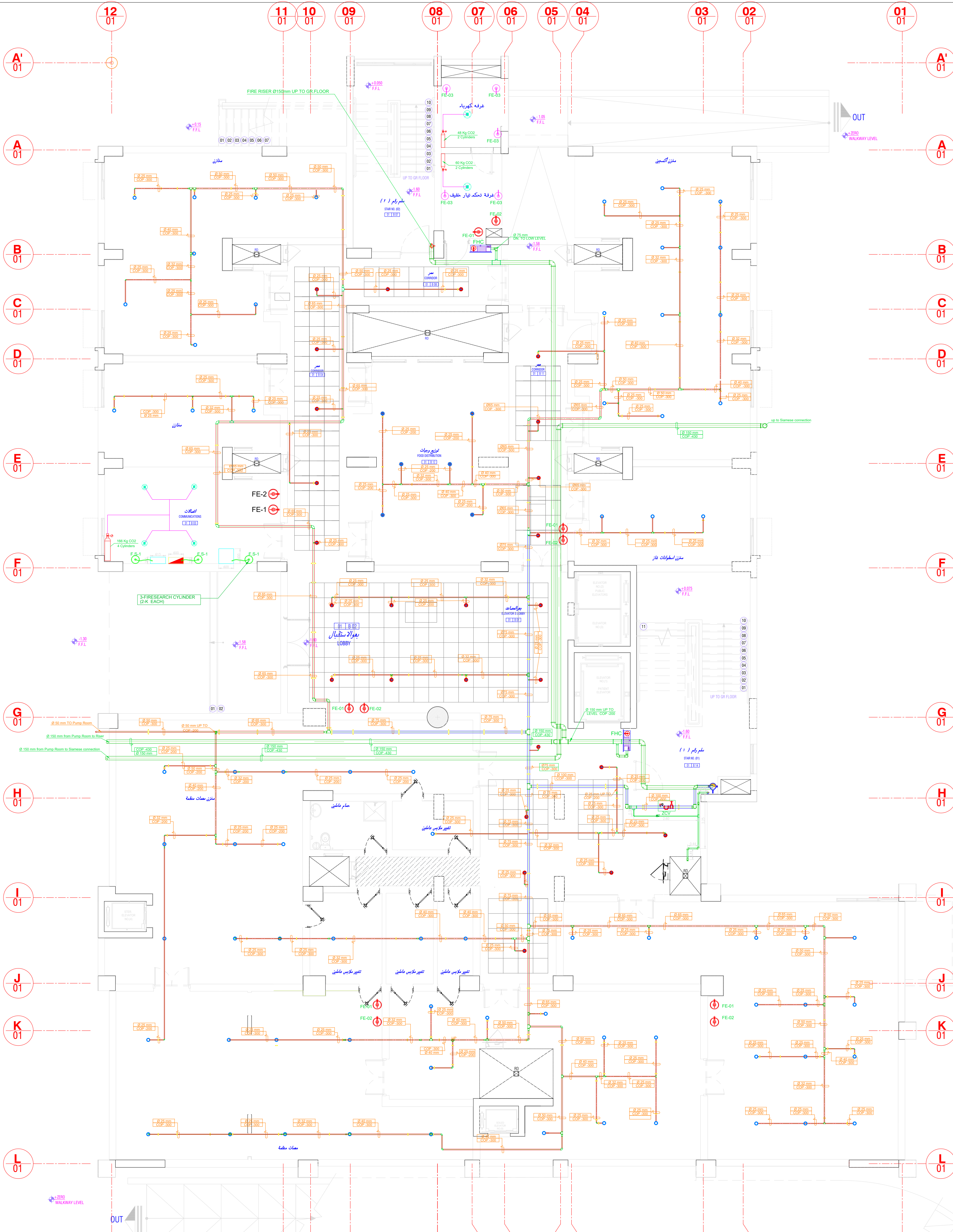
اسم المشروع

مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والأعصاب

اعداد	Prepared
احمد عاشور محمود عليان حسام محمد عبدالرحمن طه عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقى	:
مراجعة	ام.د/حموده محمد دردير

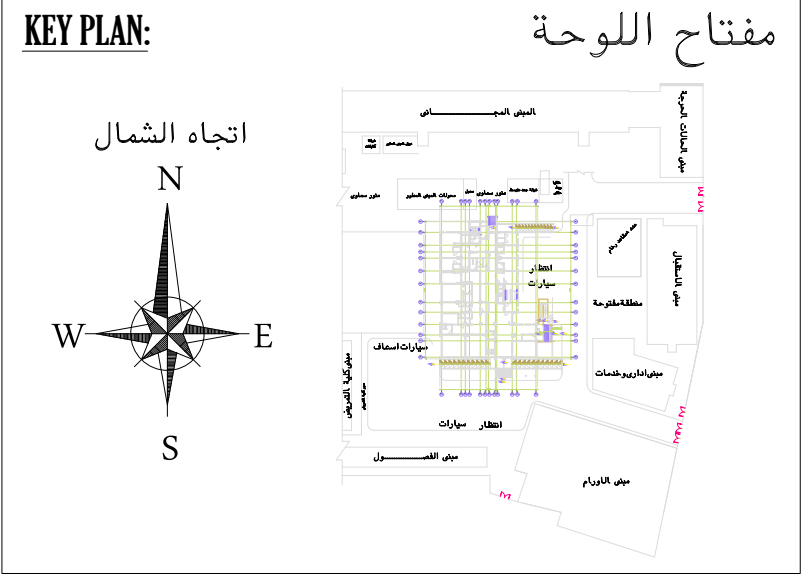
كلية الهندسة-جامعة جنوب الوادي

List of Drawings for Fire Fighting Works		
NO.	Sheet Name	Scale
100	Basement Floor Plan	1:100
101	Ground Floor Plan	
102	First Floor Plan	
103	Second Floor Plan	
104	Third Floor Plan	
105	Fourth Floor Plan	
106	Fifth Floor Plan	
107	Sixth Floor Plan	
108	Seventh Floor Plan	
109	Roof Plan	
110	Pump Room	--
111	Details	



SYMBOL & LEGEND	
SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (5kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

- ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
- ALL WORKS SHOWN ON DRAWINGS ARE COORDINATED WITH HVAC WORKS.
- ALL WORKS ARE COORDINATED BASED ON ARCH PLANS, CEILINGS AND STRUCTURE.
- ANY DEVIATIONS FROM DESIGN DRAWING ARE DUE TO COORDINATION WITH OTHER TRADES AND BUILDER WORK.
- ALL LEVELS ARE MEASURED FROM BOTTOM OF CONCRETE SLAB.
- ALL EQUIPMENT AND MATERIAL ARE AS PER APPROVED SUBMITTAL.
- ALL PIPE LEVEL IS MEASURED FROM CENTER LINE OF PIPE UNLESS OTHERWISE NOTED.
- ALL DETAILS TO BE SUBMITTED IN SEPARATE DRAWING ACCORDING TO PRELIMINARY APPROVAL AND SUBMITTALS.
- FIRE STAND PIPE RISERS & ABOVE GROUND FIRE PIPES
- MATERIAL ARE SEAMLESS BLACK STEEL PIPE.
- SPRINKLERS ARE (K-FACTOR = 5.6)



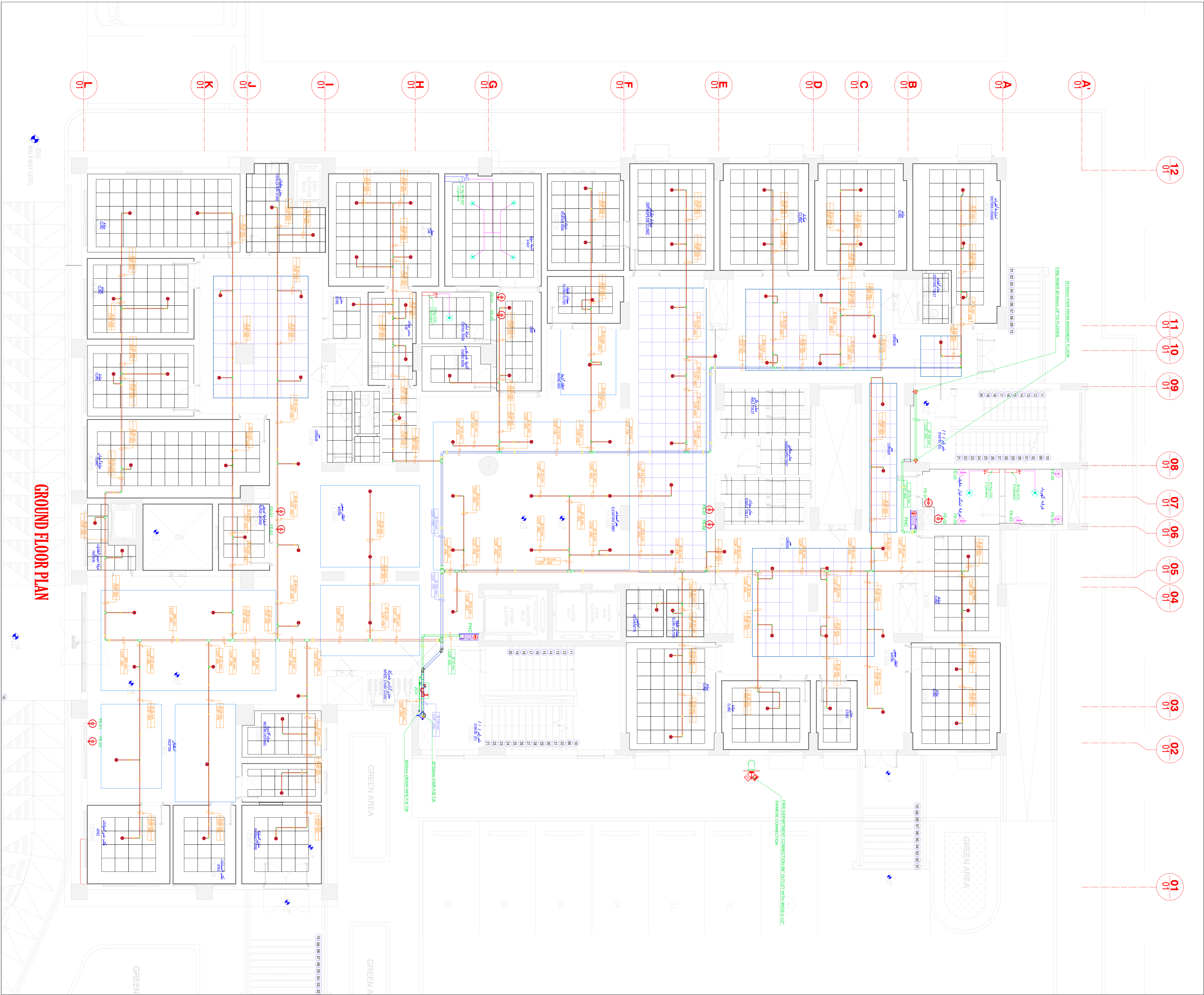
PROJECT TITLE: اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والاعصاب

SUBJECT OF DRAWING: اسم اللوحة
FIRE FIGHTING WORKS
BASEMENT FLOOR PLAN

Type:	F.F SHOP DRAWING			
Drawing No:	FF-BAS.F-100			
Date:	18-06-2025	Scale:	1 : 100	Sheet size: A3
Rev.	Date	Description		Rev. By
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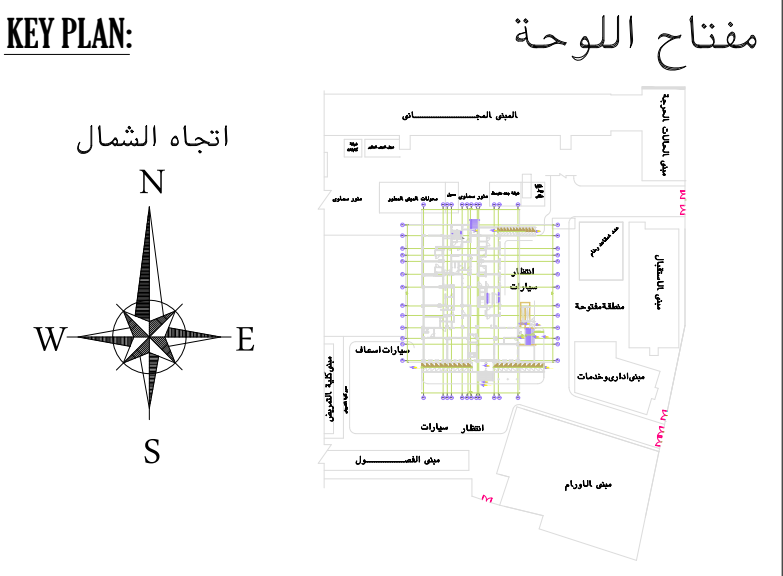
Prepared	اعداد
احمد عاشور محمود عليان حسام محمد عبدالرحمن طه عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي	
مراجعة ام.د.حموده محمد دردير	

MANAGER APPROVAL	الاعتمادات
مراجعة ام.د.حموده محمد دردير	
المكتب الفني كلية الهندسة-جامعة جنوب الوادي	



SYMBOL & LEGEND	
SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (5kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

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- MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTOR = 5.6)



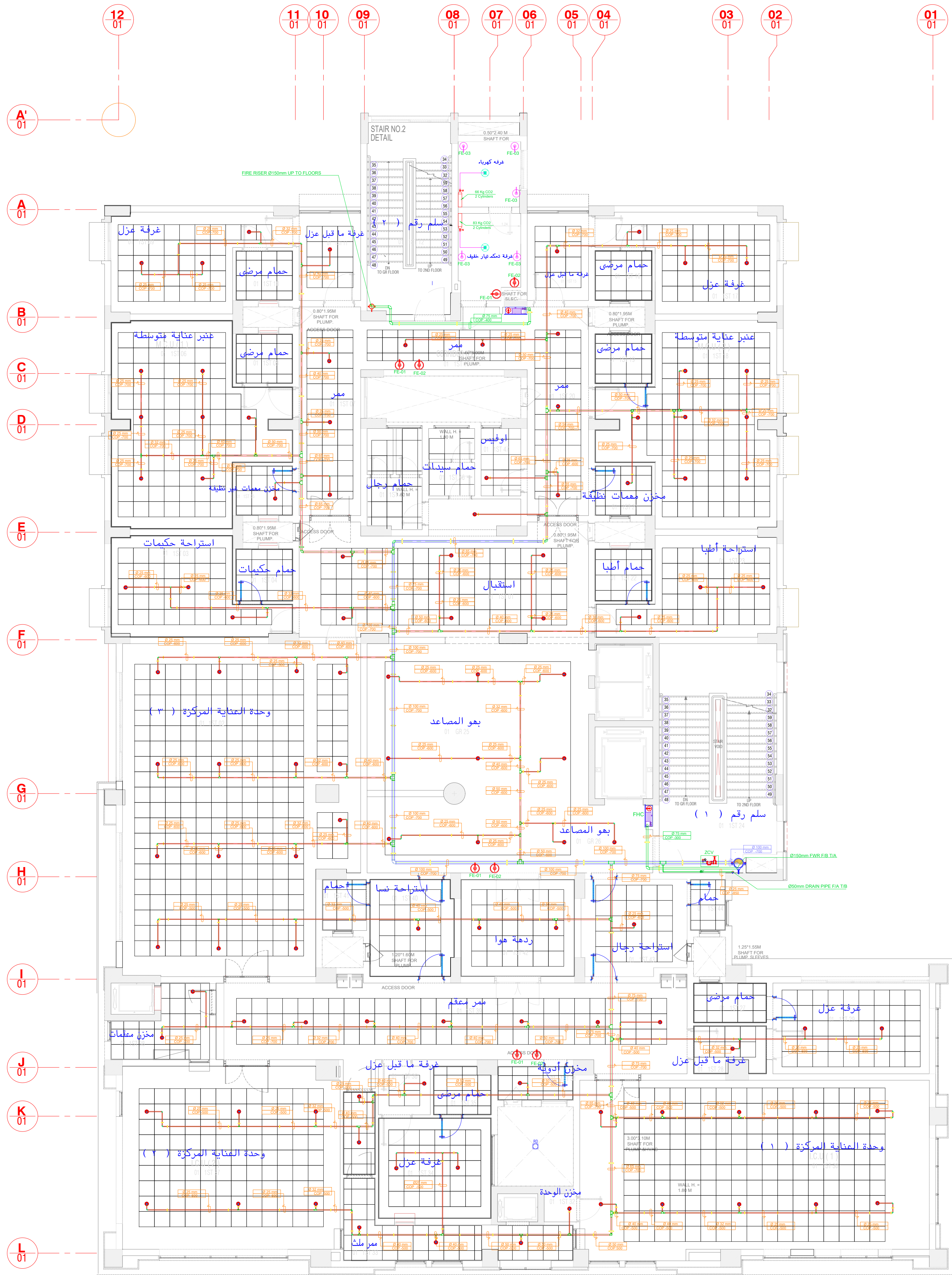
PROJECT TITLE: اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والأعصاب

SUBJECT OF DRAWING : اسم اللوحة:
FIRE FIGHTING WORKS
GROUND FLOOR PLAN

Type:	F.F SHOP DRAWING		
Drawing No:	FF-GRF-101		
Date:	18-06-2025	Scale:	1 : 100
Rev.	Date	Description	Rev. By
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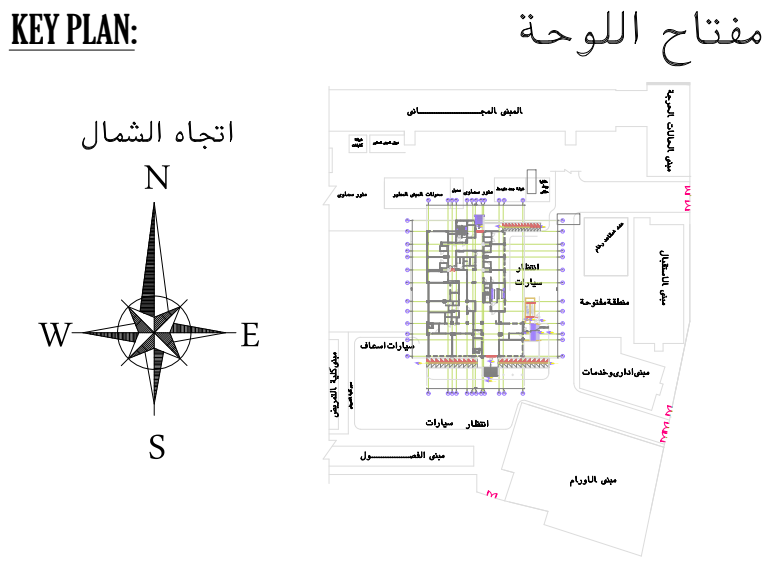
Prepared	اعداد
	احمد عاشور محمود عليان حسام محمد عبدالرحمن طه عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقى
	مراجعة : ام/د/حموده محمد دردير

MANAGER APPROVAL	الاعتمادات
	مراجعة : ام/د/حموده محمد دردير
	المكتب الفني كلية الهندسة-جامعة جنوب الوادي



SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	FE-1 PORTABLE DRY POWDER FIRE EXTINGUISHER (8kg)
	FE-2 PORTABLE FIRE EXTINGUISHER (CO2)
	FE-3 FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

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- FIRE STAND PIPE RISERS & ABOVE GROUND FIRE PIPES
- MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTORE = 5.6)



اسم المشروع
اسم المستشفى
اسم الطوارئ
اسم الجراحات القلب والصدر والمخ والاعصاب

اسم اللوحة:

FIRE FIGHTING WORKS
FIRST FLOOR PLAN

Type:	F.F SHOP DRAWING			
Drawing No:	FF-F1-102			
Date: 18/6/2025	Scale: 1 : 100	Sheet size:	A3	
Rev.	Date	Description	Rev. By	
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02				
03				
04				
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	مراجعة امد/حموده محمد دردير

MANAGER APPROVAL	الاعتمادات
	مراجعة امد/حموده محمد دردير
	المكتب الفني كلية الهندسة-جامعة جنوب الوادى

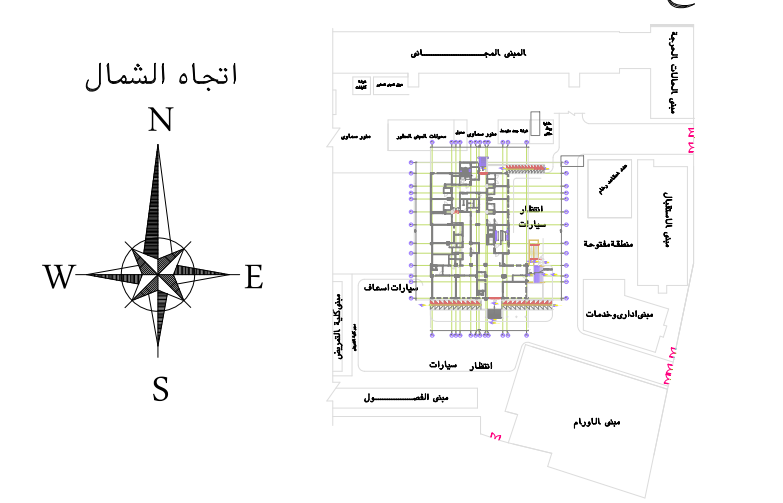


SYMBOL & LEGEND

SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (8kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

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- FIRE STAND PIPE RISERS & ABOVE GROUND FIRE PIPES
- MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTORE = 5.6)

KEY PLAN:



اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والاعصاب

اسم اللوحة:
FIRE FIGHTING WORKS
SECOND FLOOR PLAN

Type:	F.F SHOP DRAWING			
Drawing No:	FF-F2-103			
Date:	18-06-2025	Scale:	1 : 100	Sheet size: A3
Rev.	Date	Description	Rev. By	
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03				
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	احمد عاشور محمود عليان حسام محمد عبدالرحمن طه عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي
	مراجعة
	ام-د/حموده محمد دردير

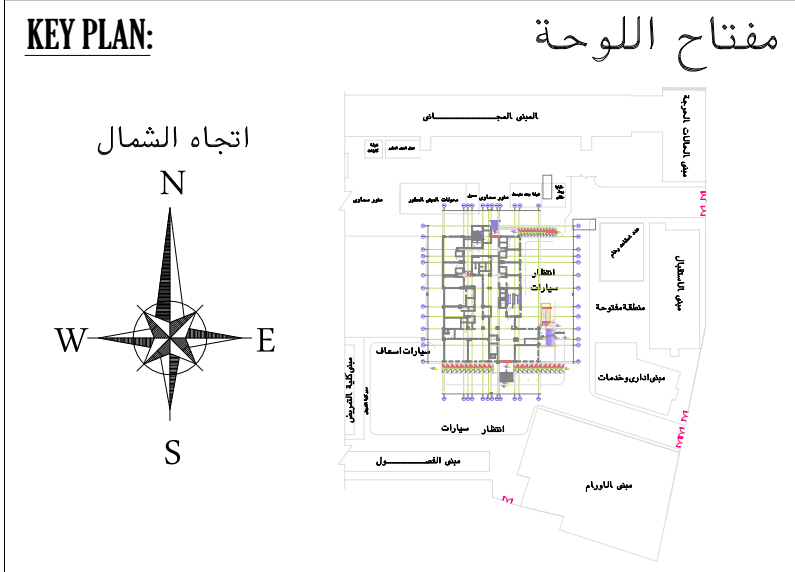
MANAGER APPROVAL	الاعتمادات
	مراجعة
	ام-د/حموده محمد دردير
	المكتب الفني كلية الهندسة-جامعة جنوب الوادي



THIRD FLOOR PLAN

SYMBOL & LEGEND	
SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (8kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

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9. FIRE STAND PIPE RISERS & ABOVE GROUND FIRE PIPES
10. MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTOR = 5.6)



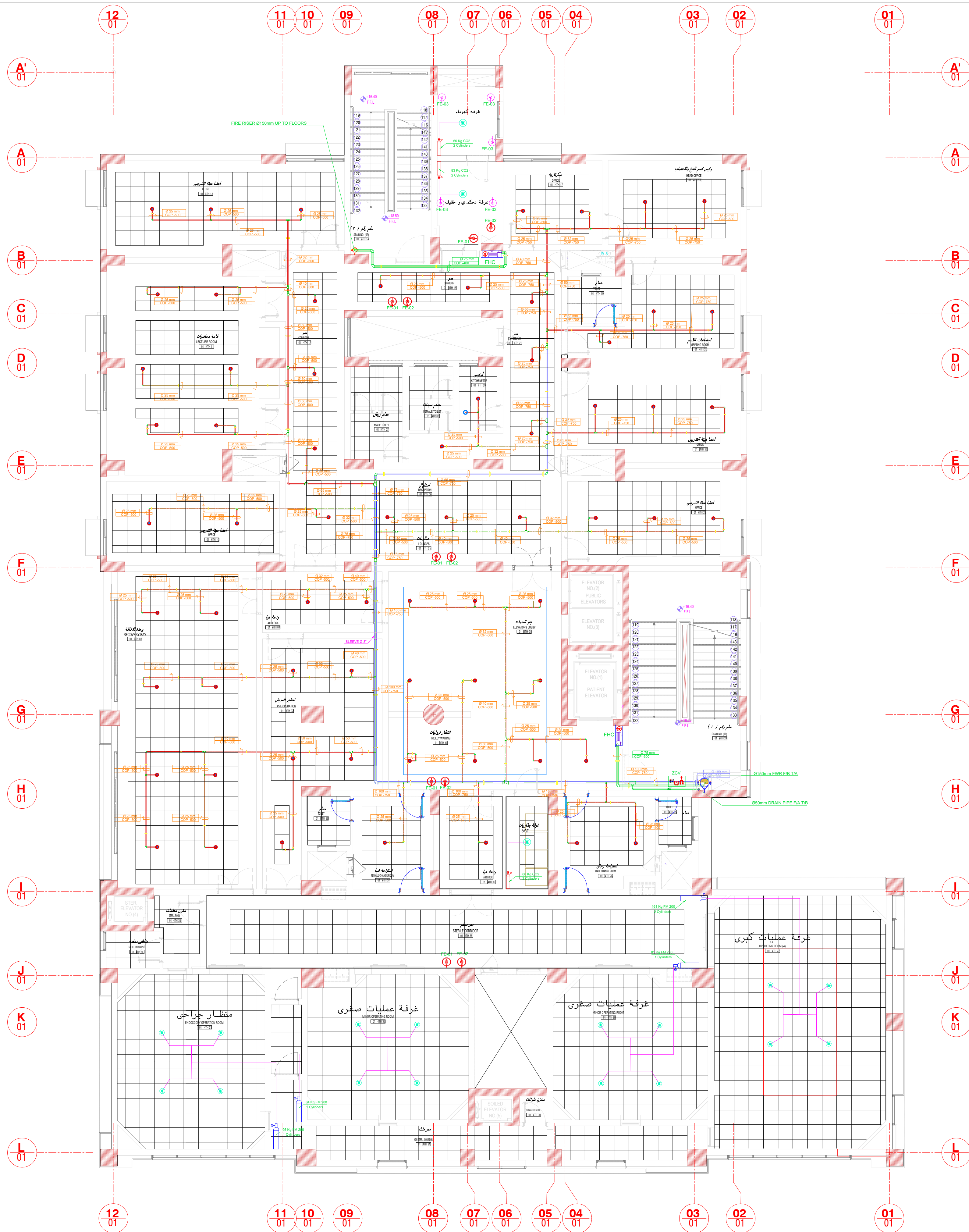
PROJECT TITLE: اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والاعصاب

SUBJECT OF DRAWING: اسم اللوحة:
FIRE FIGHTING WORKS
THIRD FLOOR PLAN

Type:	F.F SHOP DRAWING		
Drawing No:	FF-F3-104		
Date:	18-06-2025	Scale:	1 : 100
Rev.	Date	Description	Rev. By
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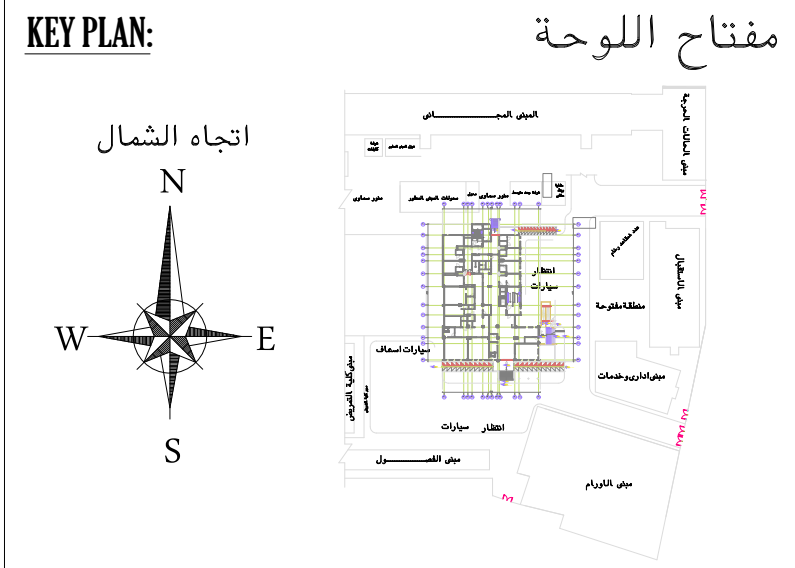
Prepared	اعداد
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	مراجعة
	ام-د/حموده محمد دردير

MANAGER APPROVAL	الاعتمادات
	مراجعة
	ام-د/حموده محمد دردير
	المكتب الفني كلية الهندسة-جامعة جنوب الوادي



SYMBOL & LEGEND	
SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (5kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

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9. FIRE STAND PIPE RISERS & ABOVE GROUND FIRE PIPES
10. MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTOR = 5.6)



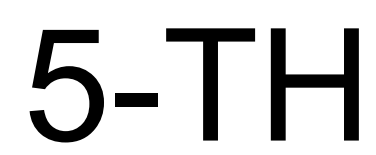
PROJECT TITLE : اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والاعصاب

SUBJECT OF DRAWING : اسم اللوحة
FIRE FIGHTING WORKS
FOURTH FLOOR PLAN

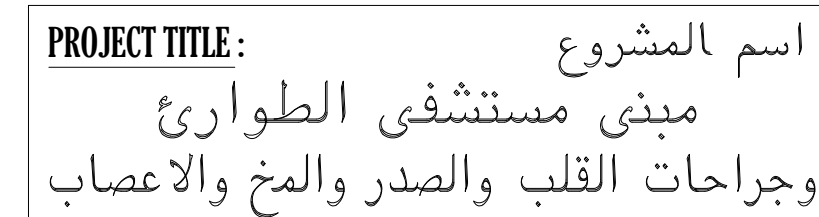
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Rev.	Date	Description	Rev. By
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03			
04			
05			

Prepared	اعداد
	احمد عاشور محمود عليان حسام محمد عبدالرحمن طه عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي
	مراجعة
	ام-د/حموده محمد دردير

MANAGER APPROVAL	الاعتمادات
	مراجعة
	ام-د/حموده محمد دردير
	المكتب الفني كلية الهندسة-جامعة جنوب الوادي

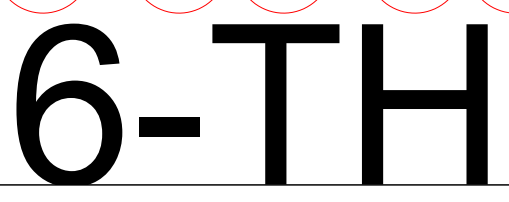


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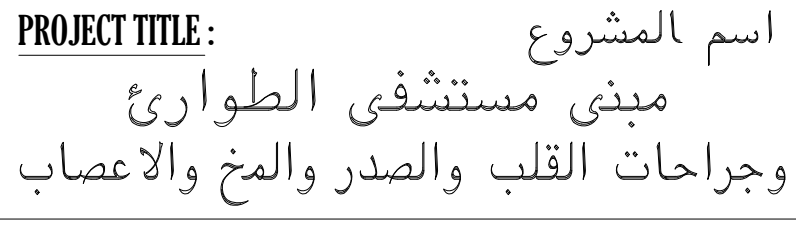


Prepared	اعداد
	احمد عاشور محمود عليان حسام محمد عبدالرحمن طله عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي
	مراجعة
	ام/د/حموده محمد دردير

الاعتمادات	MANAGER APPROVAL
مراجعة	ام.د.حموده محمد دردير
المكتب الفني	كلية الهندسة- جامعة جنوب الوادي



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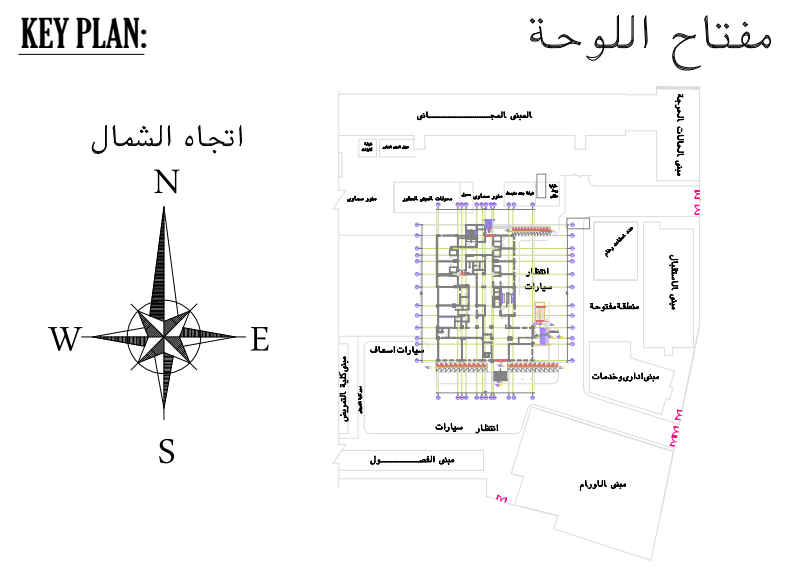
اعداد	Prepared
احمد عاشور محمود عليان حسام محمد عبدالرحمن طله عبدالحفيظ سباعي محمود حمدي عرابي محمود عاطف عبدالباقي	:
مراجعة	ام د/ حموده محمد دردير

الاعتمادات	MANAGER APPROVAL
مراجعة	ام.د. حموده محمد دردير
المكتب الفني	كلية الهندسة- جامعة جنوب الوادي



SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (8kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
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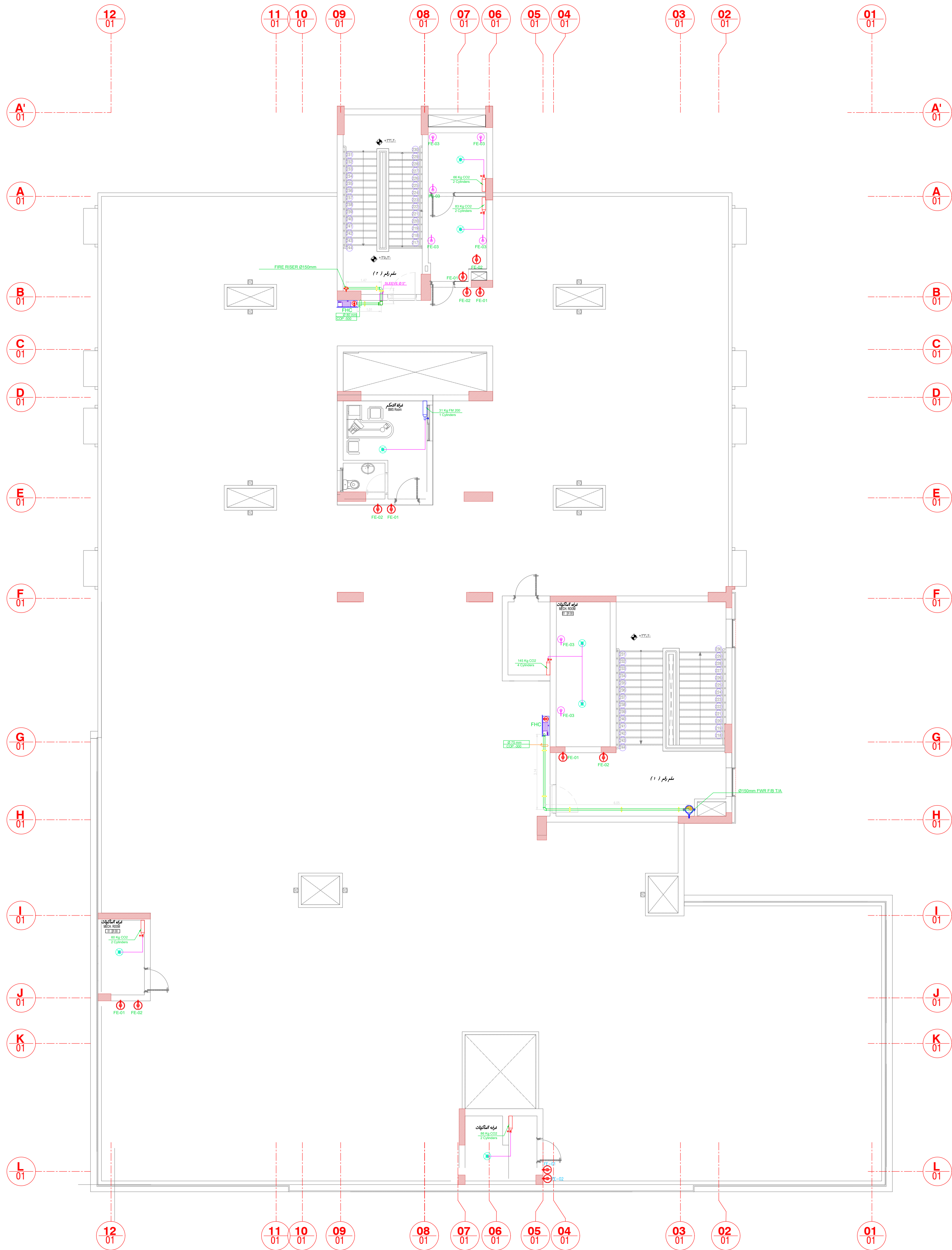
PROJECT TITLE : اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والاعصاب

SUBJECT OF DRAWING : اسم اللوحة:
FIRE FIGHTING WORKS
SEVENTH FLOOR PLAN

Type:	F.F SHOP DRAWING		
Drawing No:	FF-F7-108		
Date:	18-06-2025	Scale:	1 : 100
Rev.	Date	Description	Rev. By
01			
02			
03			
04			
05			

Prepared	اعداد
احمد عاشور محمود عليان حسام محمد عبدالرحمن طله عبدالحيث سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي	
ام-د/حموده محمد دردير	مراجعة

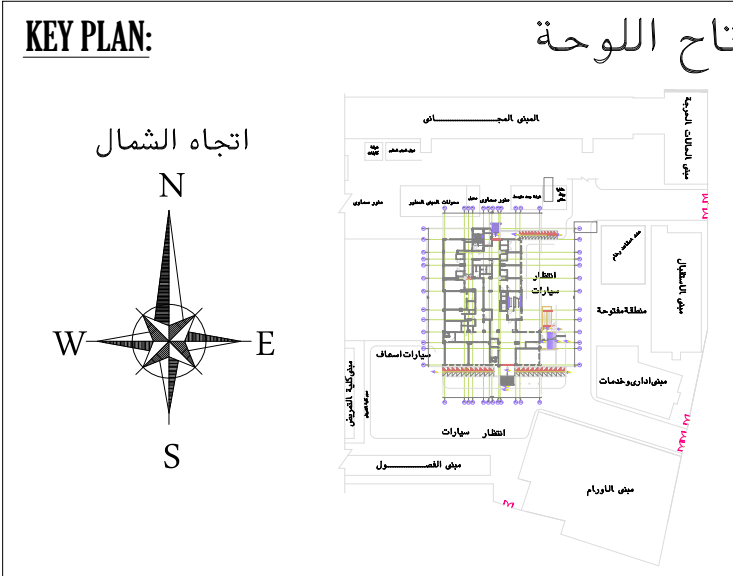
MANAGER APPROVAL	الاعتمادات
ام-د/حموده محمد دردير	مراجعة
المكتب الفني	كلية الهندسة-جامعة جنوب الوادي



ROOF PLAN

SYMBOL & LEGEND	
SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (6kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

- ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
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- FIRE STAND PIPE RISERS & ABOVE GROUND FIRE PIPES
- MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTORE = 5.6)



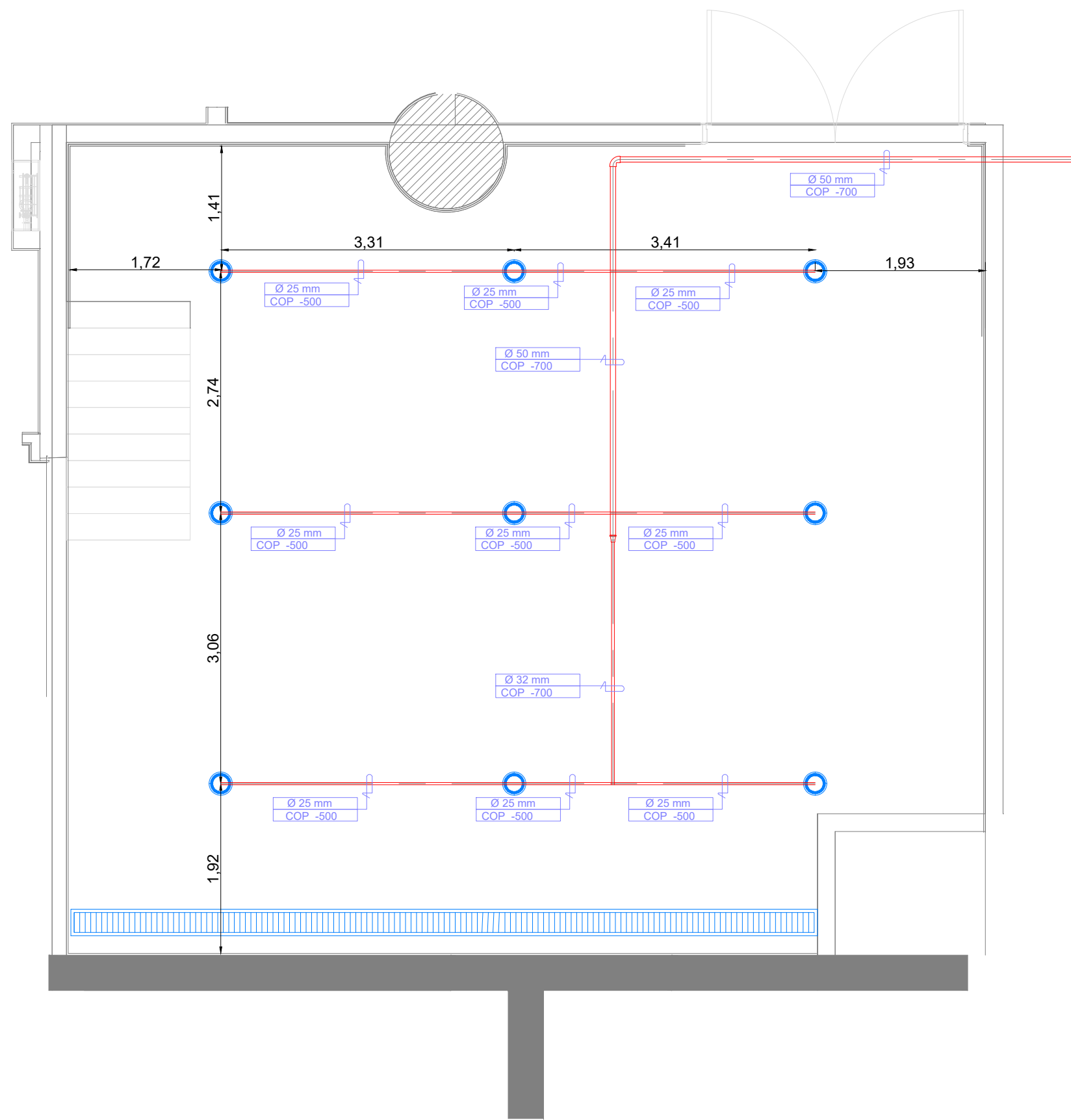
PROJECT TITLE : اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والاعصاب

SUBJECT OF DRAWING : اسم اللوحة:
FIRE FIGHTING WORKS
ROOF PLAN

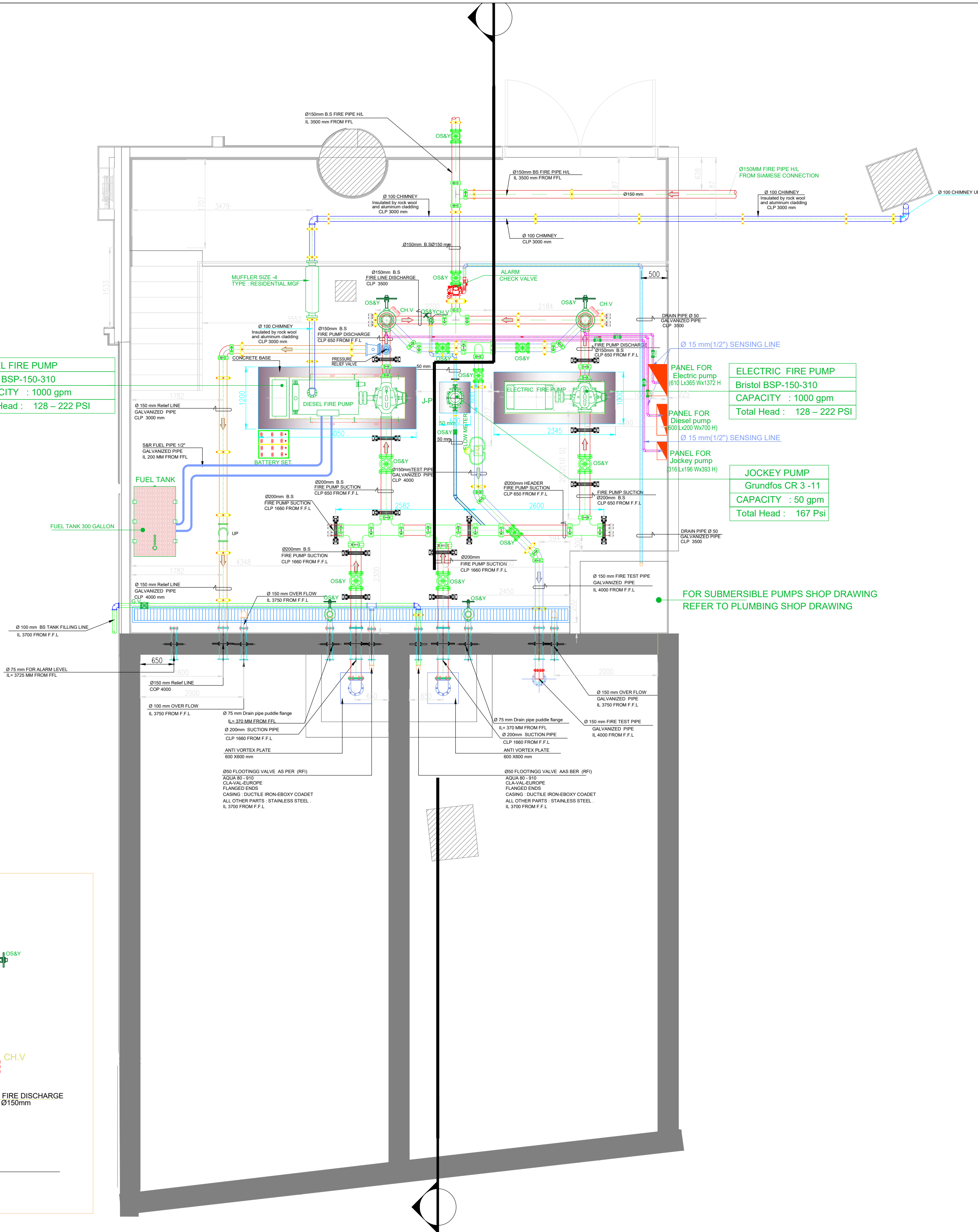
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Drawing No:	FF-RF-F-109		
Date:	18-06-2025	Scale:	1 : 100
Rev.	Date	Description	Rev. By
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02			
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Prepared	اعداد
	احمد عاشور محمود عليان حسام محمد عبدالرحمن طه عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي
	مراجعة ام-د/حموده محمد دردير

MANAGER APPROVAL	الاعتمادات
	مراجعة ام-د/حموده محمد دردير
	المكتب الفني كلية الهندسة-جامعة جنوب الوادي



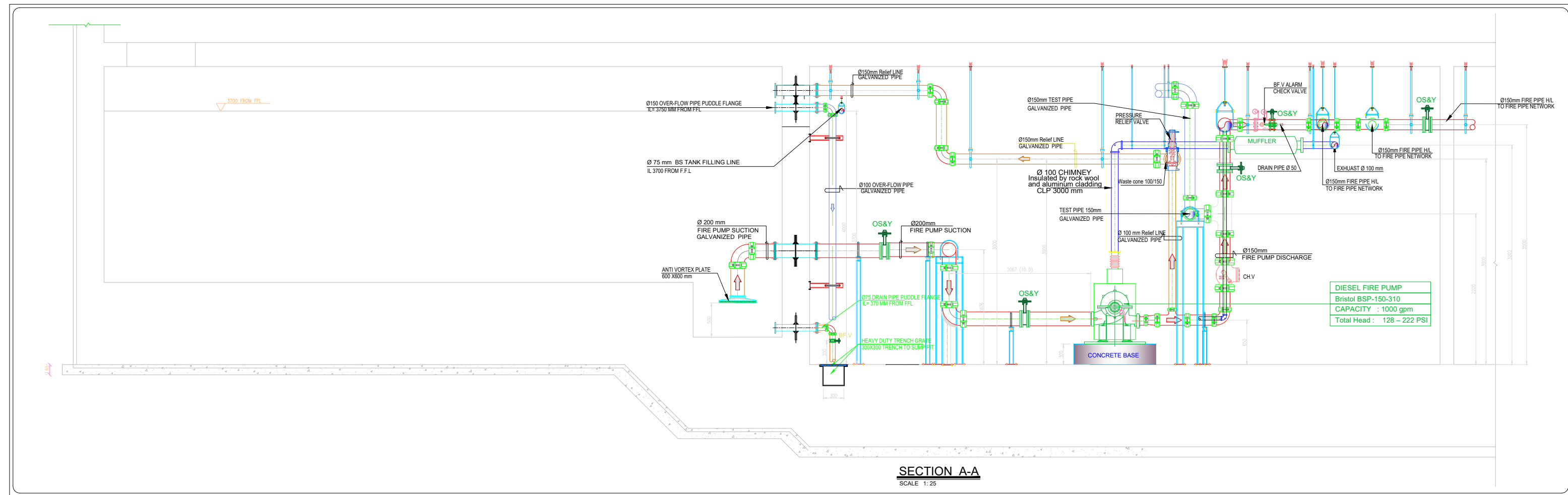
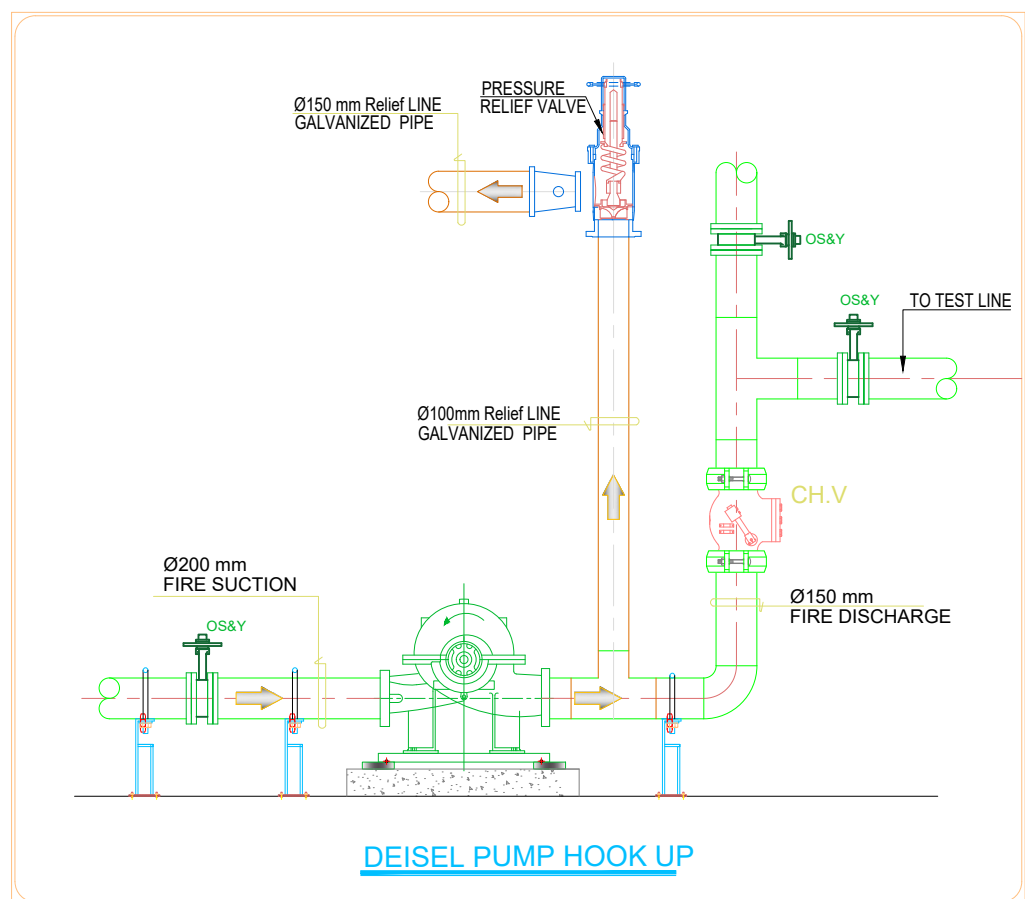
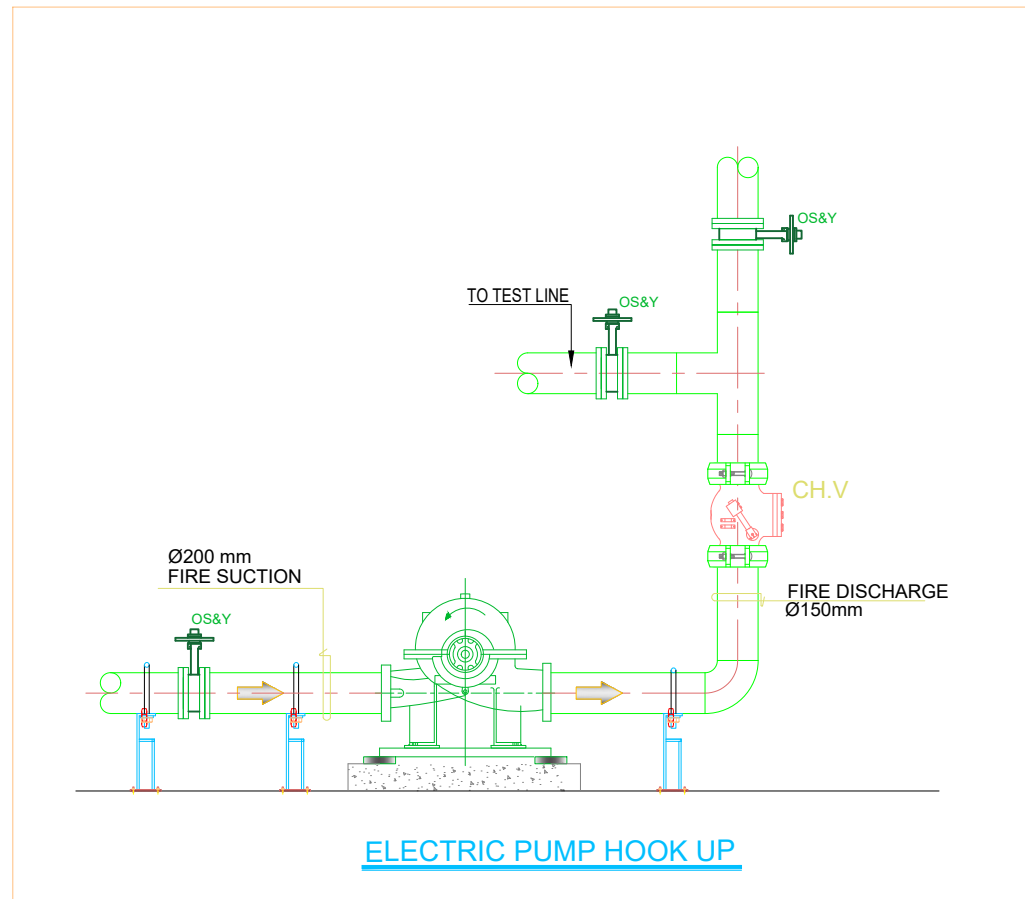
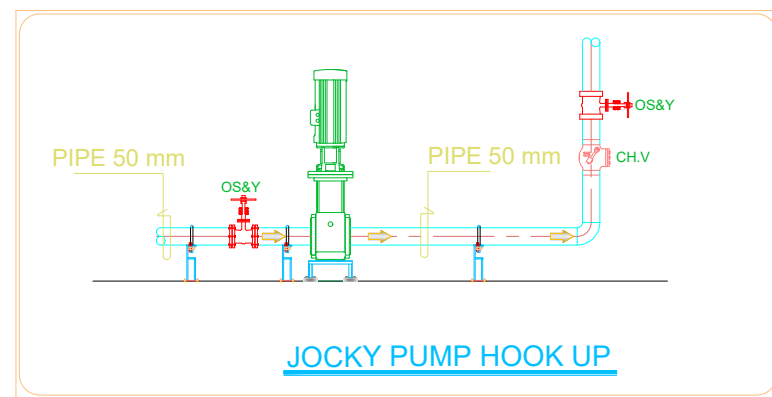
DIESEL FIRE PUMP
Bristol BSP-150-310
CAPACITY : 1000 gpm
Total Head : 128 – 222 PSI



FOR SUBMERSIBLE PUMPS SHOP DRAWING
REFER TO PLUMBING SHOP DRAWING

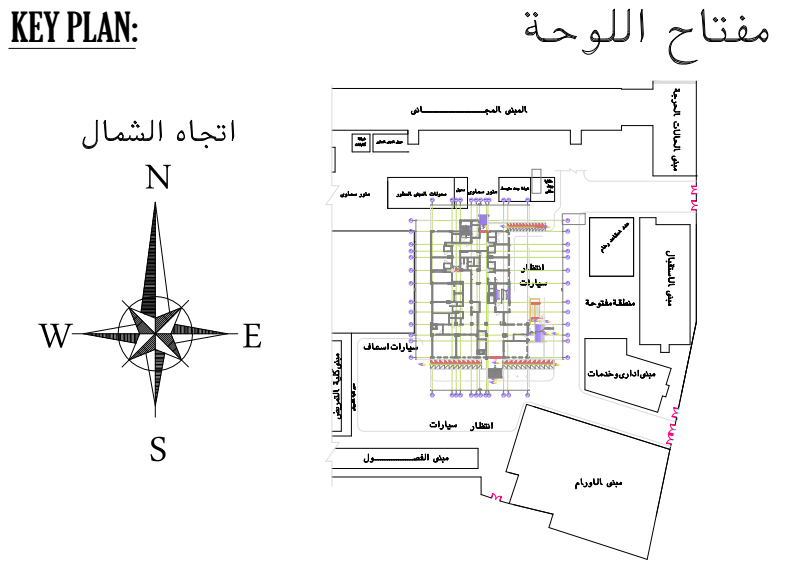
ELECTRIC FIRE PUMP
Bristol BSP-150-310
CAPACITY : 1000 gpm
Total Head : 128 – 222 PSI

JOCKEY PUMP
Grundfos CR 3 -11
CAPACITY : 50 gpm
Total Head : 167 Psi



SYMBOL & LEGEND	
SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (5kg)
	PORTABLE FIRE EXTINGUISHER (CO2)
	FIRE TERRACE
	LANDING VALVE
	CHECK VALVE
	OS&Y GATE VALVE
	DRAIN VALVE (BALL VALVE)
	PIPE RISER
	FM - 200 GAS CYLINDER
	CO2 GAS CYLINDER
	GAS SYSTEM NOZZEL

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- MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTOR = 5.6)



PROJECT TITLE: اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والأعصاب

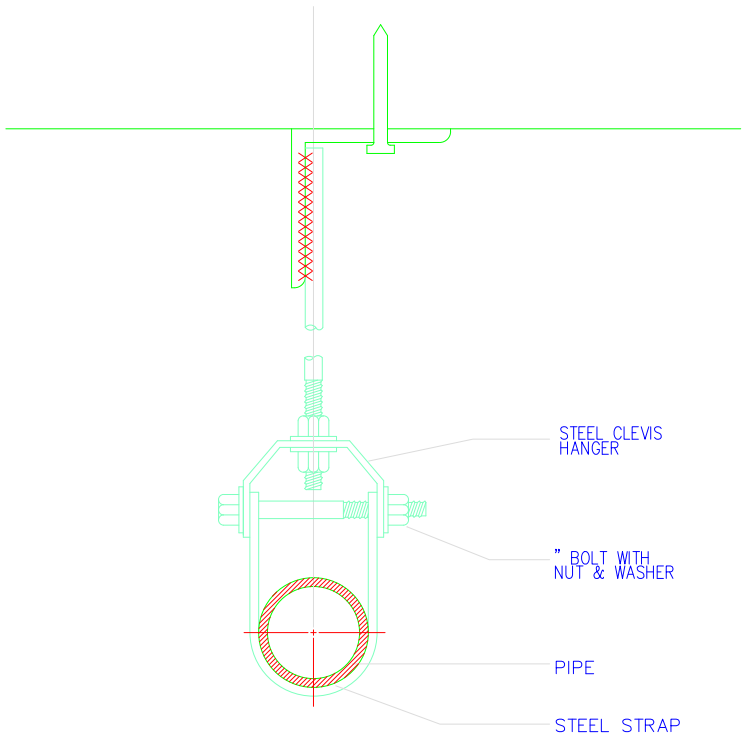
SUBJECT OF DRAWING: اسم اللوحة:
Pump Room

Type:	F.F SHOP DRAWING		
Drawing No:	FF-PR-110		
Date:	18-06-2025	Scale:	Sheet size: A3
Rev.	Date	Description	Rev. By
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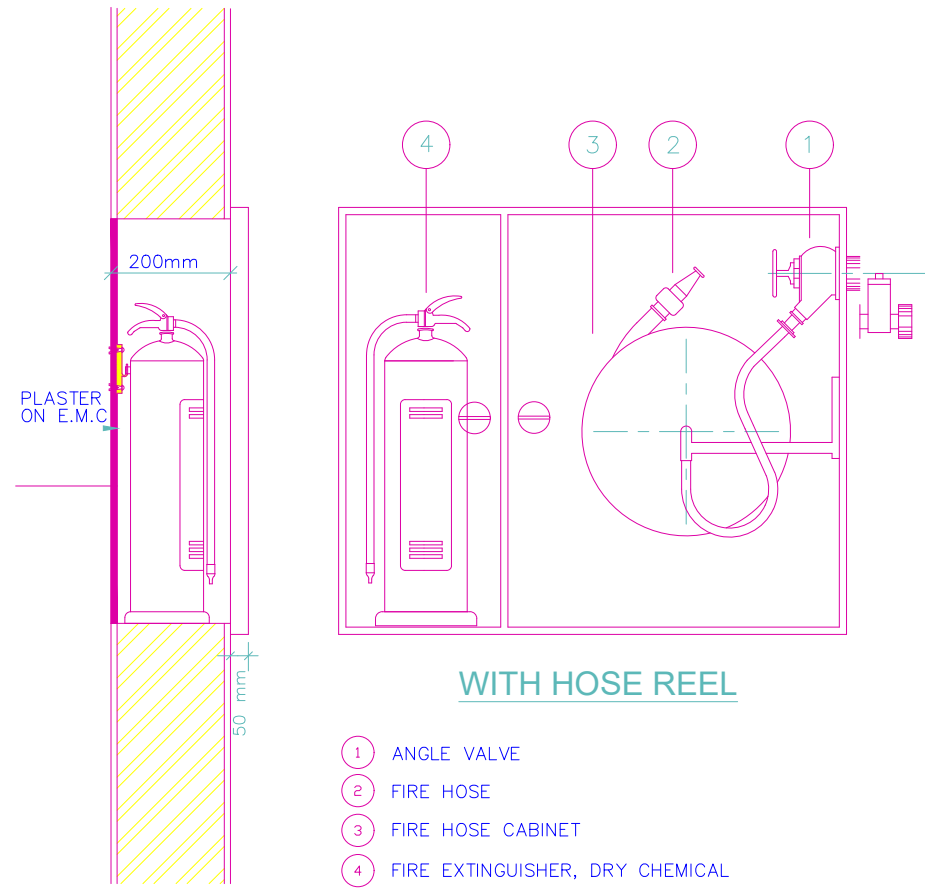
Prepared	اعداد
احمد عاشور محمود عليان حسام محمد عبدالرحمن طله عبدالحيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي	
مراجعة	ام-د/حموده محمد دردير

MANAGER APPROVAL	الاعتمادات
ام-د/حموده محمد دردير	مراجعة
المكتب الفني كلية الهندسة-جامعة جنوب الوادي	

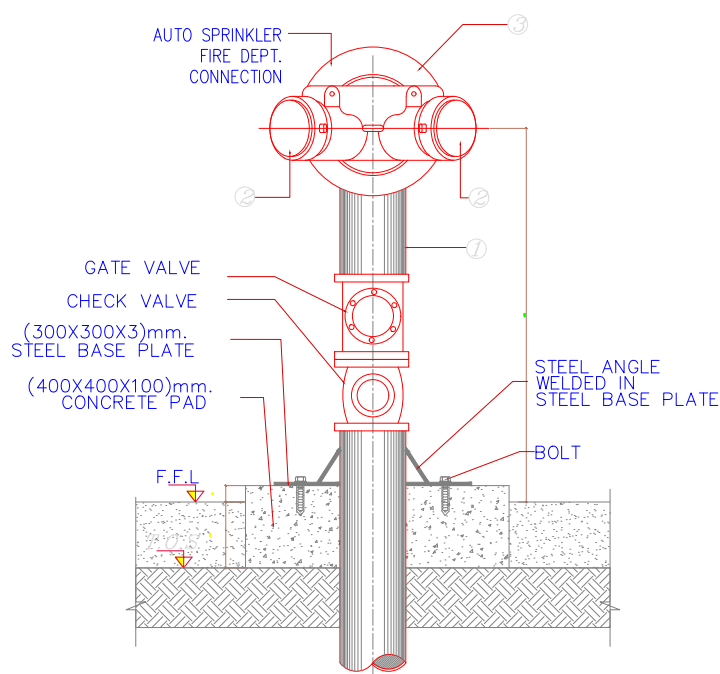
TABLE Ordinary Hazard Pipe Schedule - NFPA Of Sprinklers	
Ø1" (25)	2 Sprinklers
1 1/4" (32)	3 Sprinklers
1 1/2" (40)	5 Sprinklers
2" (50)	10 Sprinklers
2 1/2" (65)	20 Sprinklers
3" (75)	40 Sprinklers
4" (100)	100 Sprinklers



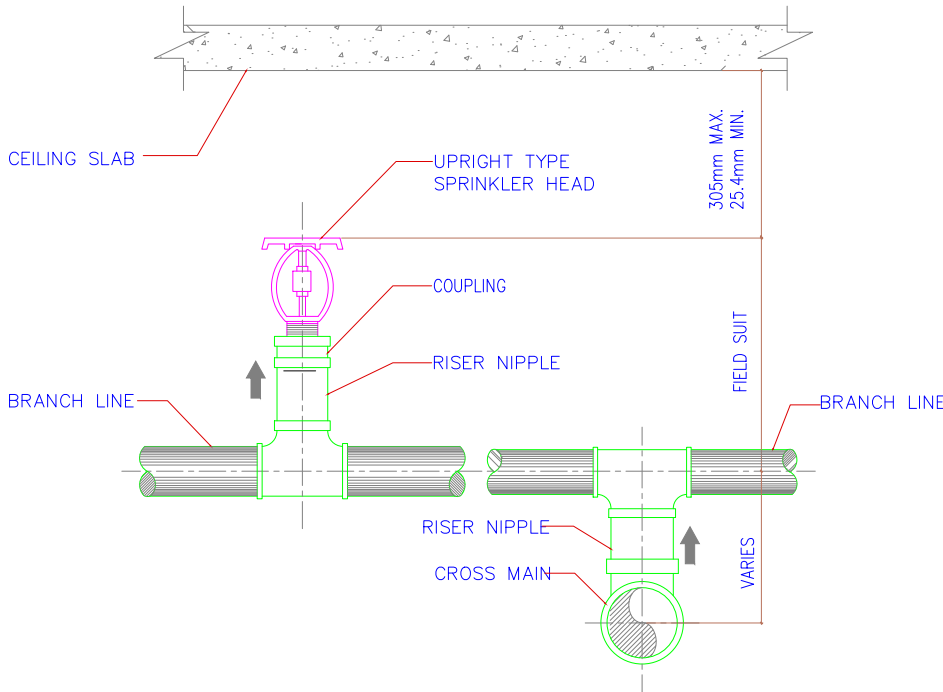
PIPE HANGER DETAIL



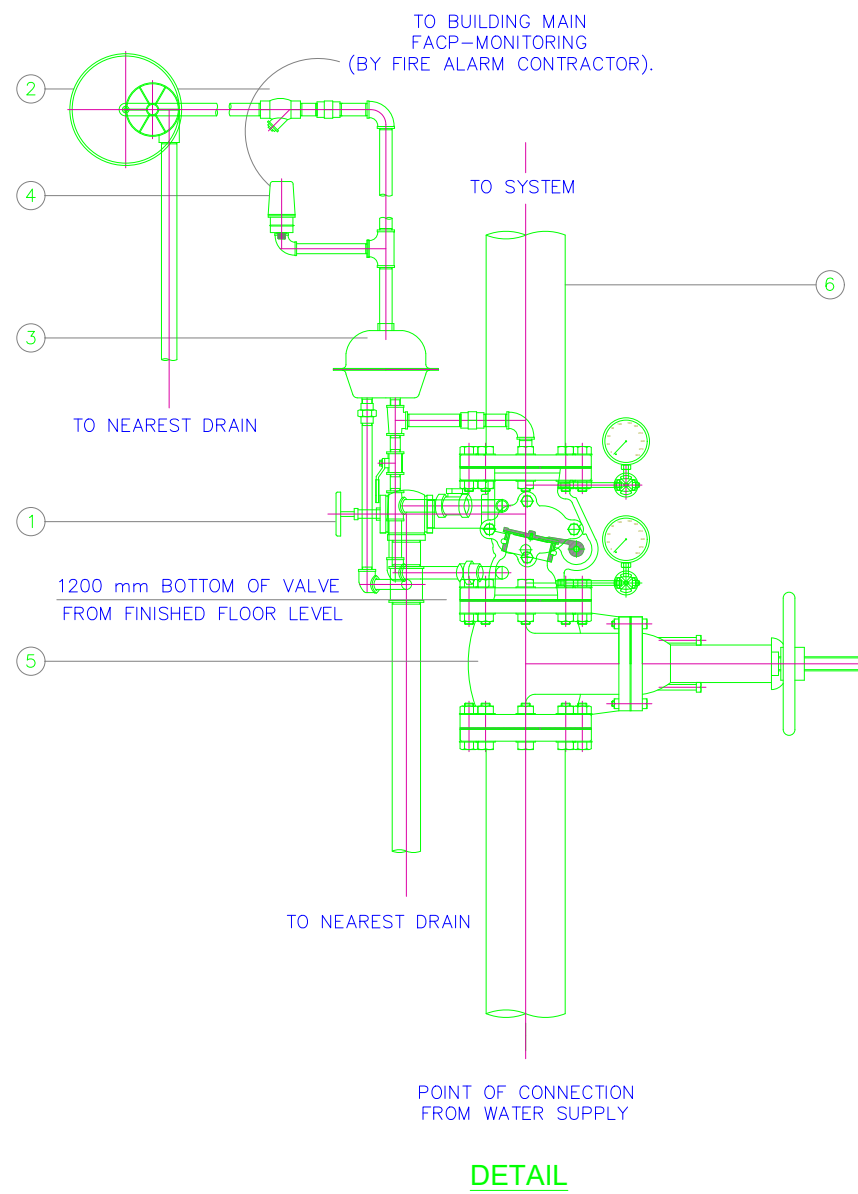
FIRE HOSE CABINET WITH EXTINGUISHER



FIRE DEPARTMENT CONNECTION DETAIL



UPRIGHT TYPE SPRINKLER DETAIL



SPECIFICATION:

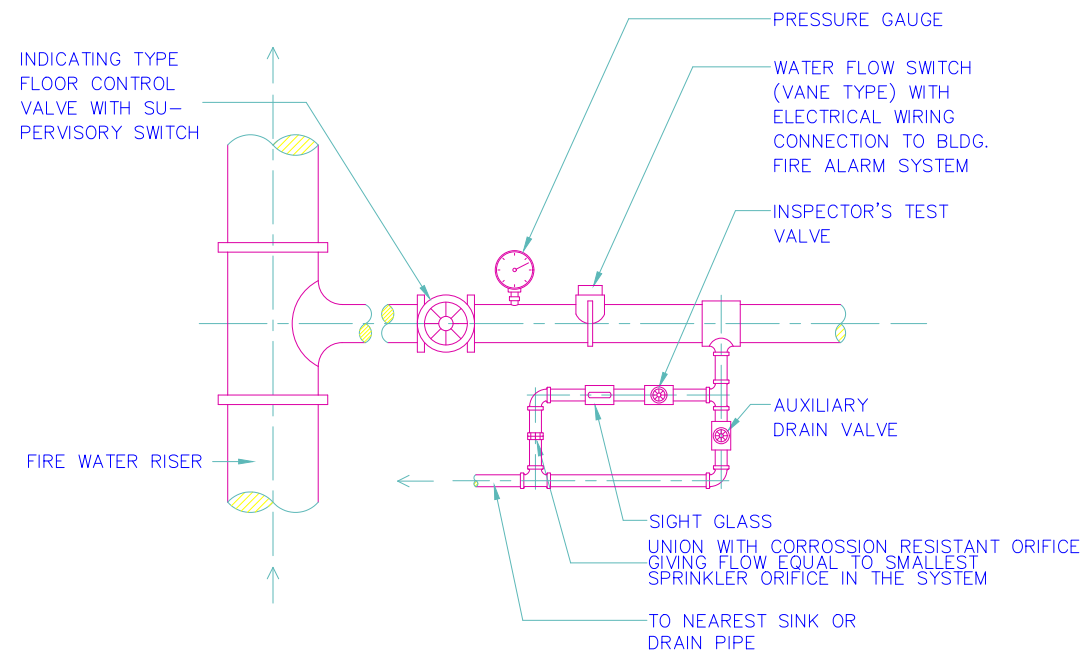
1. WET ALARM CHECK VALVE — 8" FLANGE ENDS HORIZONTAL INSTALLATION DUCTILE IRON BODY.
2. WATER MOTOR ALARM GONG — (AS PER ALARM CHECK VALVE MANUFACTURERS STANDARD PRODUCT).
3. RETARD CHAMBER — (AS PER ALARM CHECK VALVE MANUFACTURERS STANDARD PRODUCT).
4. PRESSURE ALARM SWITCH — SPDT.
5. O.S. & Y. GATE VALVE — (NORMALLY OPEN) DUCTILE IRON BODY 175 PSI MINIMUM WATER WORKING PRESSURE.
6. WET AUTO SPRINKLER — STD. SCH-40 BLACK IRON STEEL PIPE, ERW, CONFORMING WITH ASTM-A 53.

NOTES:

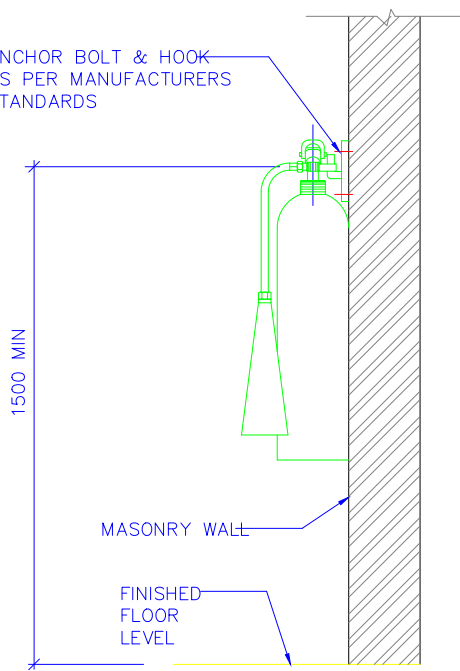
PRESSURE GAUGES, DRAIN VALVE AND OTHER MISCELLANEOUS TRIM VALVES ARE PART OF THE WATER ALARM CHECK VALVE APPURTENANCE TRIM PACKAGE.

PORTABLE FIRE EXTINGUISHER MOUNTING DETAILS

ALARM CHECK VALVE ASSEMBLY



FLOOR CONTROL VALVE ASSEMBLY

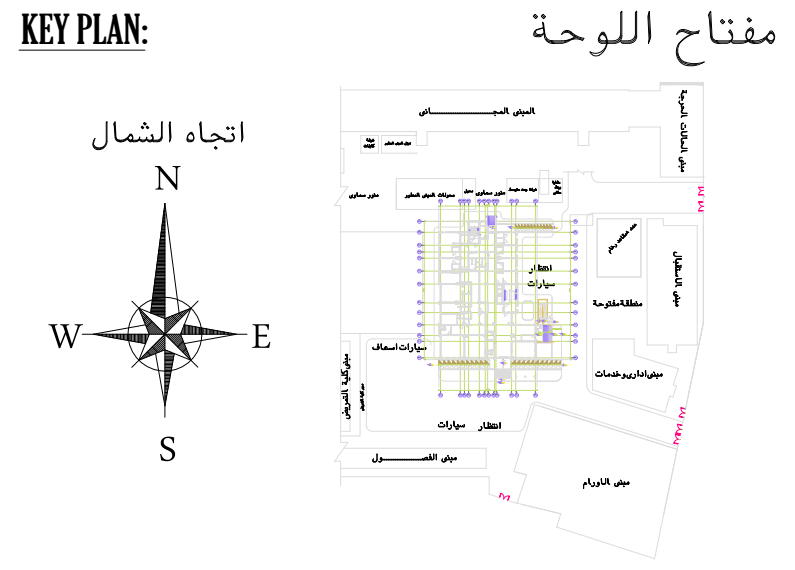


SYMBOL & LEGEND

SYMBOL	DESCRIPTION
	PENDANT SPRINKLER
	SIDE WALL SPRINKLER
	FIRE HOSE CABINET
	PORTABLE DRY POWDER FIRE EXTINGUISHER (5kg.)
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10. MATERIAL ARE SEAMLESS BLACK STEEL PIPE .
SPRINKLERS ARE (K-FACTORE = 5.6)

KEY PLAN:



PROJECT TITLE : اسم المشروع
مبنى مستشفى الطوارئ
وجراحات القلب والصدر والمخ والأعصاب

SUBJECT OF DRAWING : اسم اللوحة:

Details

Type:	F.F SHOP DRAWING		
Drawing No:	FF-DET-111		
Date:	18-06-2025	Scale:	Sheet size: A3
Rev.	Date	Description	Rev. By
01			
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Prepared	اعداد
احمد عاشور محمود عليان حسام محمد عبدالرحمن طه عبدالحفيظ سباعي محمود حمدي محمد عرابي محمود عاطف عبدالباقي	
ام.د/حموده محمد دردير	مراجعة

MANAGER APPROVAL	الاعتمادات
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المكتب الفني	كلية الهندسة-جامعة جنوب الوادي